Tactix: Sweep Original-Core-Unit Audit

Report to Andrew

25 September 2026

Outcome

The N4 scheduling ledger contained a stale open-case description. It said Tactix visits only derived Sweep proof steps and cannot expose an original clause that becomes a root unit in an extracted core. Source inspection shows that commit `7a668e76` (12 September) added ordered clausal-core export, original-row eligibility filtering, inline unit commitment, and a one-step proof for an original clause. The ledger now says *implemented; acceptance open*. No solver semantics changed in this audit.

Evidence and limits

The September 7 isolated candidate matched the seed-563 boundary: original core clause 44 exposed unit -970 at query 20, the 94 emitted units matched in order, and the original-unit native LRAT fixture checked independently. The candidate's core-only Rank-34 runtime worsened, so that isolated result was not a performance promotion. The current source retains the core-capture mechanism in `src/rules/sweep/sweep.cpp`. The broader September 12 online-refactor fuzz and matched gates exercised the committed candidate, but do not substitute for a current-source regression that specifically forces this original-core path through native LRAT and DSR.

The correction is supported by the frozen `2026-09-07-original-core-convergence.md` report, source history and the current Sweep capture/planner. Documentation size, option coverage, and progress checks pass; the six pre-existing document-size reserve warnings remain. The repository's historical reports were not edited.

Next actions

Add the exact current-source original-core-unit regression and independent proof checks, then run the cumulative correctness and performance gates before closing this N4 row. Continue the active Opus-owned fuzz and full Debug jobs; the H2 DSR checker contract and post-finalization clause-addition API decision remain unresolved. N1–N4 and the Rank/B+C/WV acceptance gates are not yet met. The requested full Tier-D comparison at 5,000 seconds, four CPUs and 32 GiB on hwulab remains deferred until the goal is achieved.